

Yash Narayan

(352)-642-3004 | yashnarayan@ufl.edu | [linkedin.com/in/yashnara](https://www.linkedin.com/in/yashnara) | github.com/yashnaray

EDUCATION

University of Florida

Bachelor of Science, Computer Science

Honors: UF Distinguished Scholar, Dean's List

Relevant Coursework: Data Structures & Algorithms, Computer Organization, Intro to Software Engineering, Operating Systems.

Gainesville, FL

Aug. 2024 – Jun 2027

GPA: 4.0 / 4

EXPERIENCE

Undergraduate Research Assistant

June 2025 – Present

Timothy J. Garrett Laboratory | *Python, OpenCV, NumPy, SciPy, PyQt*

Gainesville, FL

- Architected modular data processing system for mass spectrometry workflows, managing **3GB+** analytical datasets with structured validation pipelines
- Optimized signal-processing routines using NumPy vectorization and SciPy statistical methods, improving detection accuracy in isomer analysis.
- Parallelized I/O-bound operations with Python multithreading and caching, reducing analysis runtime from 1.5 hours to 10 minutes (**98%** improvement).
- Built PyQt GUI with automated reporting and data validation, reducing manual debugging overhead and improving data quality controls.

Software Engineering Intern

June 2025 – Aug 2025

Rebel Pouches | *Rust, Next.js, Tailwind, MySQL*

Remote

- Engineered Rust backend services and Next.js dashboard for real-time sales/marketing analytics, enabling cross-functional teams to monitor KPIs without manual reporting.
- Translated business requirements into scalable backend modules using REST APIs and database optimization, reducing data query latency from 15 minutes to **5** seconds.
- Automated recurring data pipelines with scheduled jobs, eliminating **40+** manual reporting hours per month.
- Implemented code review process and sprint-based development, improving release stability and reducing critical post-deployment issues by majority.

PROJECTS

Market Risk Assessment

 | *Python, PyTorch, NumPy, SciPy*

Aug 2025 – Present

- Designed and trained deep learning models for financial risk profiling across large market datasets (**30GB+**).
- Implemented backtesting framework computing portfolio risk metrics (alpha, beta, Sharpe ratio, VaR).
- Built volatility estimation modules with stress-testing capabilities for investment strategy evaluation.
- Focus: Understanding machine learning application to quantitative finance and risk modeling.

LORA PCB Tester

 | *Python, Flask, Pinecone, Langchain*

Oct 2025

- Led full-stack development of validation platform for LORA radio PCB signal integrity and compliance testing
- Integrated LLM-based pipeline for schematic analysis and report generation (Markdown/PDF/DOCX), reducing manual review time from 40+ hours to under **5** minutes
- Implemented data parsing and test automation with vector database (Pinecone) for efficient schema matching
- Result: Hackathon project delivered end-to-end solution for hardware validation workflow

CLUBS & ORGANIZATIONS

- **GatorAI:** Contributed to applied machine learning projects in quantitative finance, including portfolio Value-at-Risk estimation.
- **Software Engineering Club:** Led training sessions on full-stack development, mentored cross functional teams: Led training sessions on full-stack development, mentored teams on system design and code review.
- **Open-Source Club, Florida Engineering Society, Association of Computing Machinery:** Active contributor and participant.

TECHNICAL SKILLS

Languages: C++, Python, Rust, Java, SQL, JavaScript

Frameworks & Libraries: TensorFlow, PyTorch, React, Next.js, Node.js, PyQt, OpenCV, NumPy, SciPy, REST APIs

Database & Cloud: MySQL, PostgreSQL, AWS, Docker, Kubernetes, Pinecone DB